

WEATHER FORECASTING AND ANALYSING USING PYTHON AND OPEN METEO API

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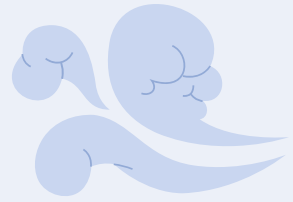
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INTRODUCTION



Greetings, respected mentors and peers! Welcome to our presentation on 'Weather Forecasting and Analysis using Python and Open-Meteo API'.

- Weather affects our daily lives — from travel and agriculture to outdoor planning and safety.
- Accurate forecasting helps individuals and businesses make better decisions.
- Technology and meteorology together enable smarter, real-time predictions.
- By analyzing weather data from different cities, we can identify trends that support effective planning.



This project demonstrates how Python and the Open-Meteo API were used to collect, analyze, and visualize real-time weather forecasts for meaningful insights.



PROJECT SELECTION RATIONALE

- ❑ **Data Availability** : Open-Meteo API offers real-time, structured weather data, making it easy to collect and analyze without manual effort.
- ❑ **Societal Relevance** : Weather forecasting supports agriculture, transport, disaster management, and daily planning — making it highly relevant.
- ❑ **Technical Learning Opportunity** : The project involved API integration, data handling, visualization, and statistics in Python — helping build real-world skills.
- ❑ **Rising Demand for Forecasting Tools** : Many industries now rely on accurate weather insights, making forecasting skills valuable in today's data-driven world.

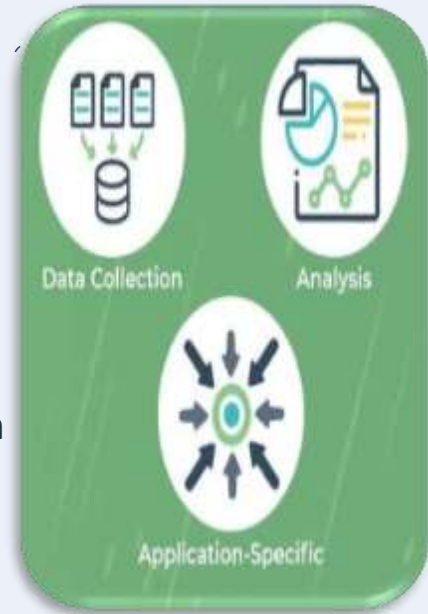


OBJECTIVES

The **primary objective** of this project is to collect, analyze, and interpret real-time weather forecast data using Python and the Open-Meteo API.

Secondary Objectives:

- ☐ **Weather Pattern Analysis** – Study temperature, weather conditions, and seasonal variations across cities.
- ☐ **Visualization of Trends** – Use line graphs and pie charts to simplify weather data.
- ☐ **Comparative City Analysis** – Compare weather behavior across major Indian cities.
- ☐ **Statistical Evaluation** – Calculate mean, median, and correlation for deeper insights.
- ☐ **Practical Applications** – Support planning in travel, agriculture, and events with weather-based decisions.



TARGET AUDIENCE



STUDENTS



RESEARCHERS



**DATA ANALYSIS
ENTHUSIASTS**



**URBAN
PLANNER**



COMMON PEOPLE

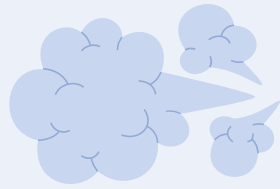
METHOD OF DATA COLLECTION



- ❑ First, we began by defining the requirement for real-time weather forecasting data.
- ❑ Following that, we identified the Open-Meteo API as a reliable and freely available source for retrieving live weather data based on city coordinates.
- ❑ Amidst various available APIs, we evaluated options based on factors like data accuracy, update frequency, accessibility, and ease of integration.
- ❑ After careful evaluation, we selected the Open-Meteo API and proceeded to integrate it into our Python environment, laying the foundation for dynamic and accurate weather data collection.



METHOD OF FORECASTING



Data Collection & Processing Steps:

- 1.City Input:** User inputs the city name through a Python script.
- 2.Geocoding:** Fetch latitude and longitude using a Geocoding API.
- 3.Data Retrieval:** Use the Open-Meteo Forecast API to obtain hourly weather data, including temperature and conditions.
- 4.Data Cleaning:** Process and clean the retrieved data in Python to ensure accuracy.

Tools & Technologies:

- Open-Meteo API:** Provides free, real-time weather forecasts without requiring an API key.
- Python Libraries:** Utilized requests for API calls and pandas for data handling.





OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025

↔ Enter city name: kolkata

Location: Calcutta, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 34.1°C

Min Temp: 25.8°C

Condition: Rain showers

Date: 2025-04-17

Max Temp: 33.2°C

Min Temp: 26.2°C

Condition: Thunderstorm

↔ Enter city name: delhi

Location: Delhi, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 36.9°C

Min Temp: 25.0°C

Condition: Partly cloudy

Date: 2025-04-17

Max Temp: 39.1°C

Min Temp: 27.2°C

Condition: Mainly clear

↔ Enter city name: mumbai

Location: Mumbai, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 33.1°C

Min Temp: 26.7°C

Condition: Partly cloudy

Date: 2025-04-17

Max Temp: 32.3°C

Min Temp: 27.7°C

Condition: Partly cloudy



OUTCOMES



↔ Enter city name: chennai

Location: Chennai, India
Weather Forecast:

Date: 2025-04-16
Max Temp: 35.9°C
Min Temp: 28.0°C
Condition: Rain showers

Date: 2025-04-17
Max Temp: 36.5°C
Min Temp: 27.8°C
Condition: Partly cloudy

↔ Enter city name: bengaluru

Location: Bengaluru, India
Weather Forecast:

Date: 2025-04-16
Max Temp: 31.4°C
Min Temp: 21.7°C
Condition: Thunderstorm

Date: 2025-04-17
Max Temp: 31.5°C
Min Temp: 20.9°C
Condition: Rain showers

METHOD OF ANALYSIS



Once the data was prepared, we performed the following analysis:

Statistical Analysis:

- **Descriptive Statistics:** Calculated mean, median, and correlation coefficients to understand data distributions and relationships.
- **Trend Analysis:** Identified patterns over time to observe weather trends across different cities.

Data Visualization:

- **Line Graphs:** Depicted temperature trends over time for each city.
- **Pie Charts:** Illustrated the distribution of various weather conditions.

Tools Used:

- **Matplotlib:** For creating static, animated, and interactive visualizations in Python.
- **Seaborn:** Enhanced data visualization with statistical graphics.

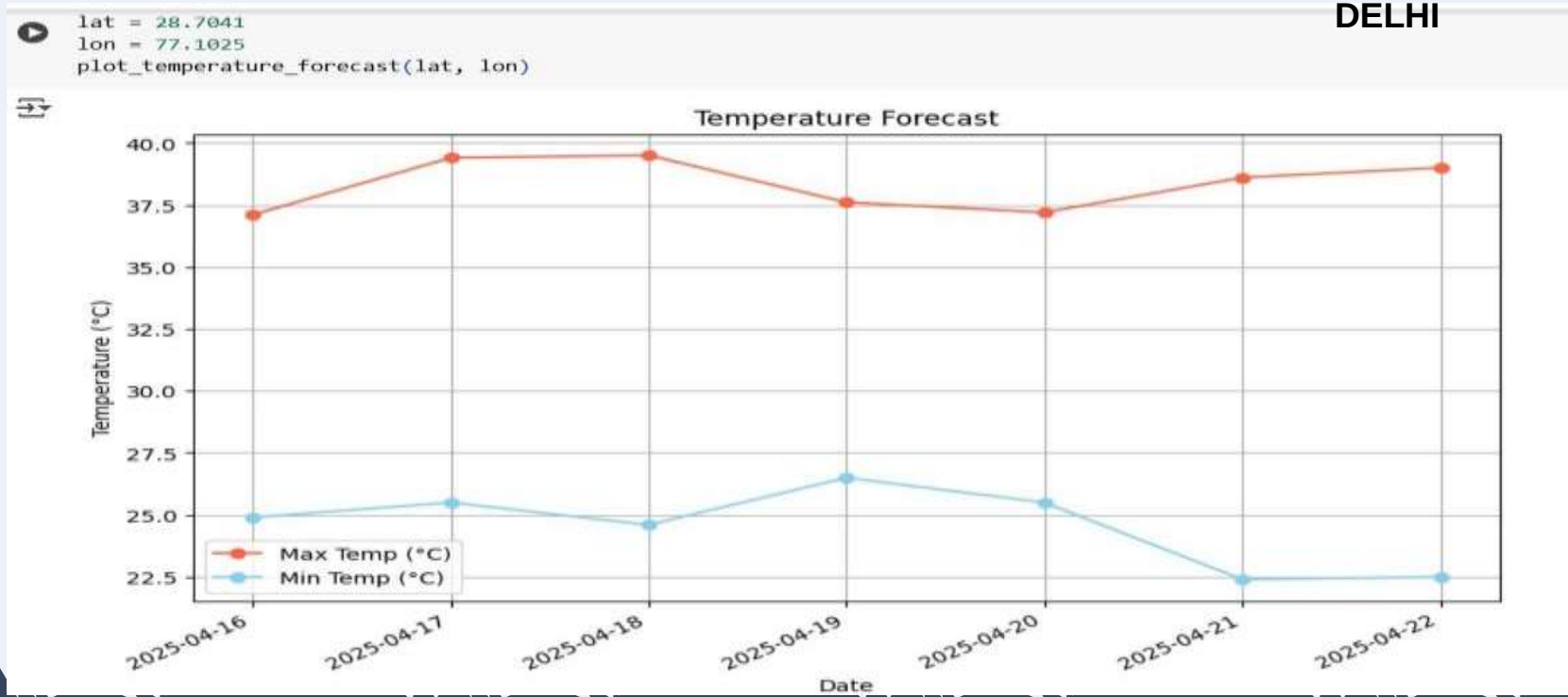




OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025

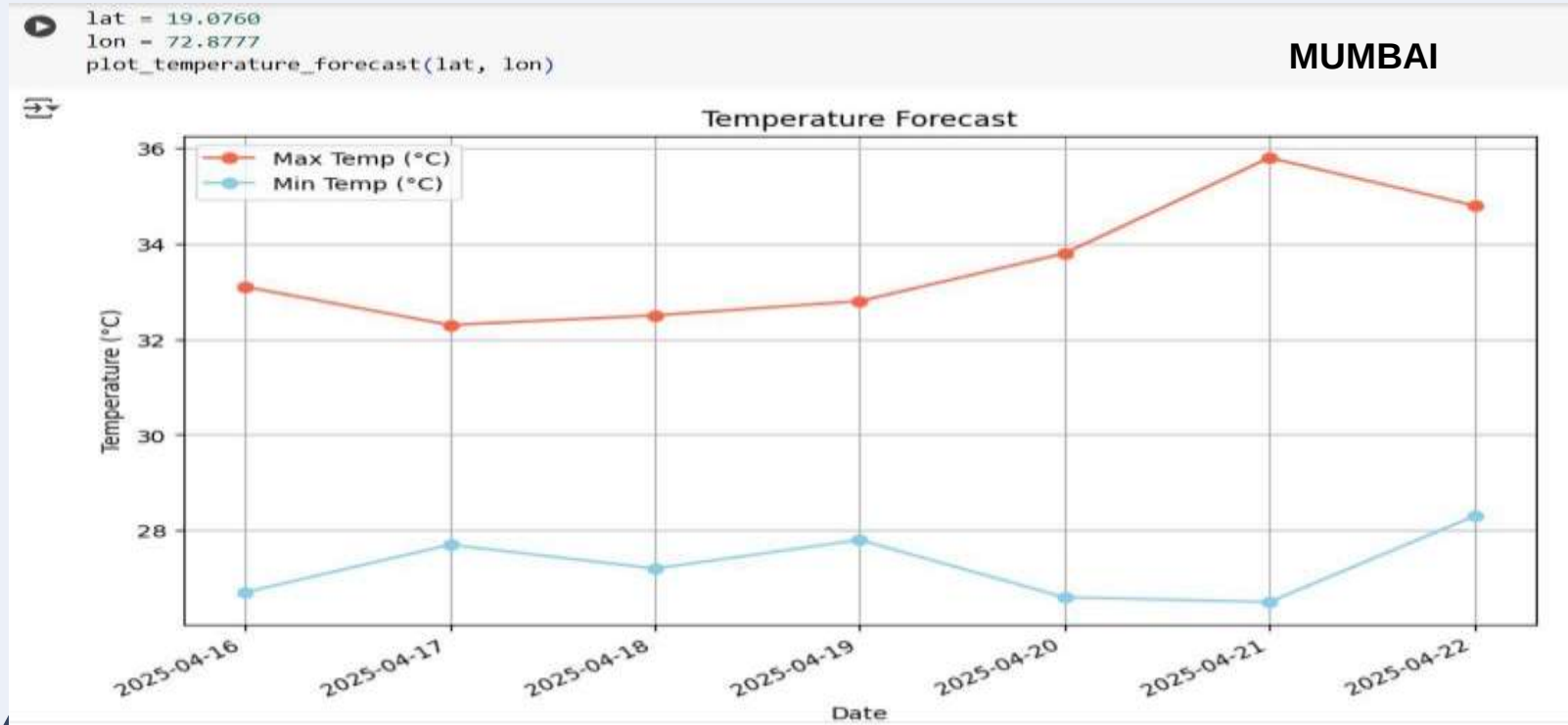




OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025





OUTCOMES

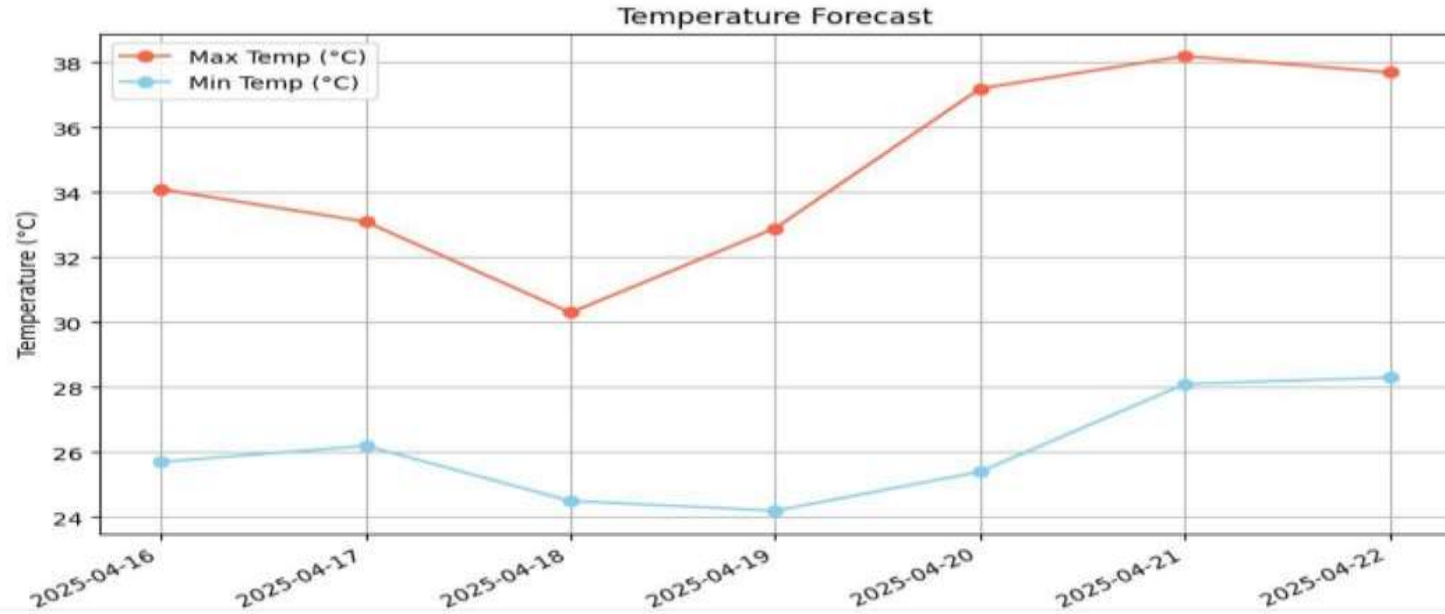


All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025



```
lat = 22.5726  
lon = 88.3639  
plot_temperature_forecast(lat, lon)
```

KOLKATA





OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025

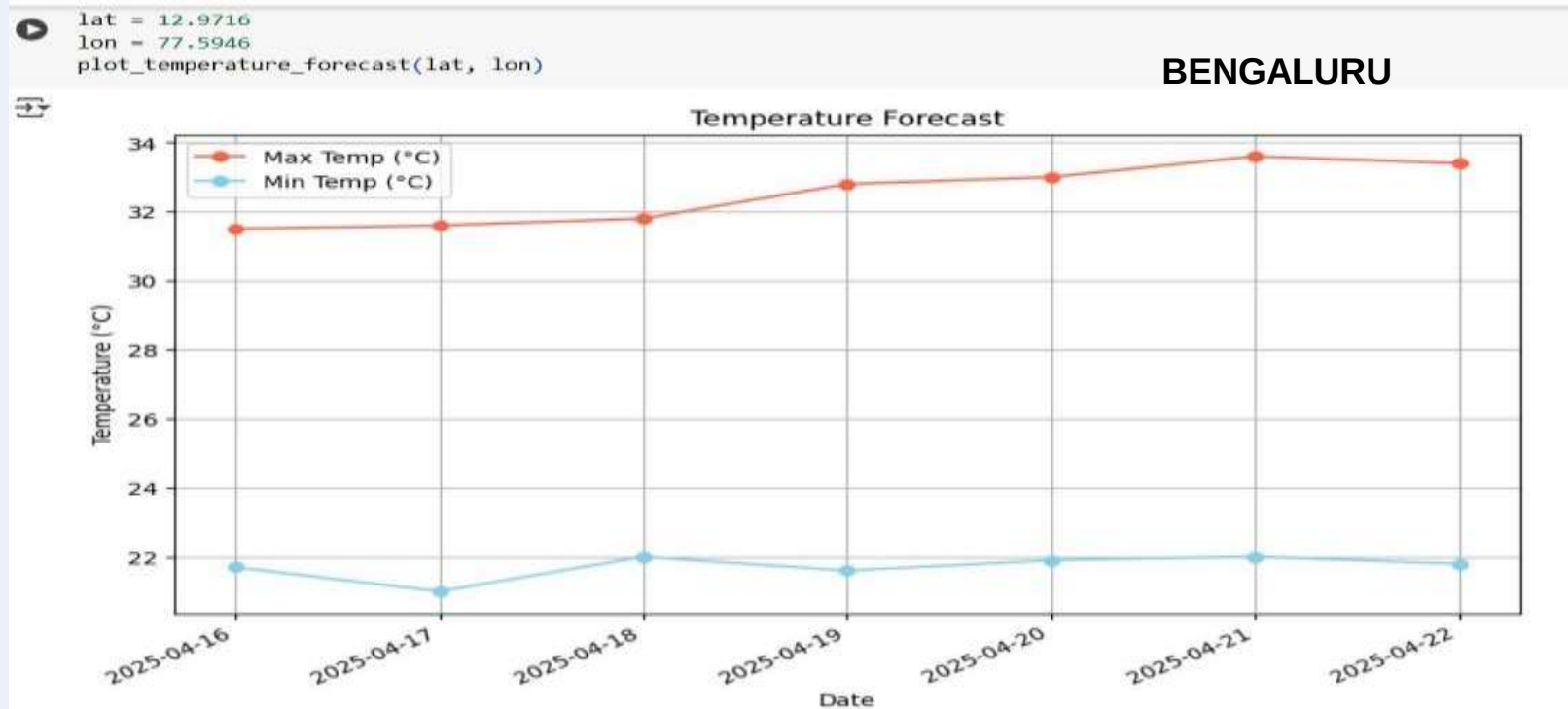




OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025





OUTCOMES

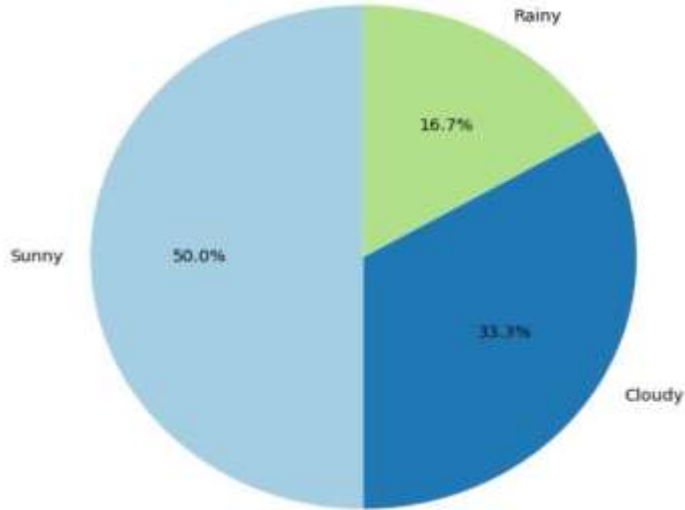


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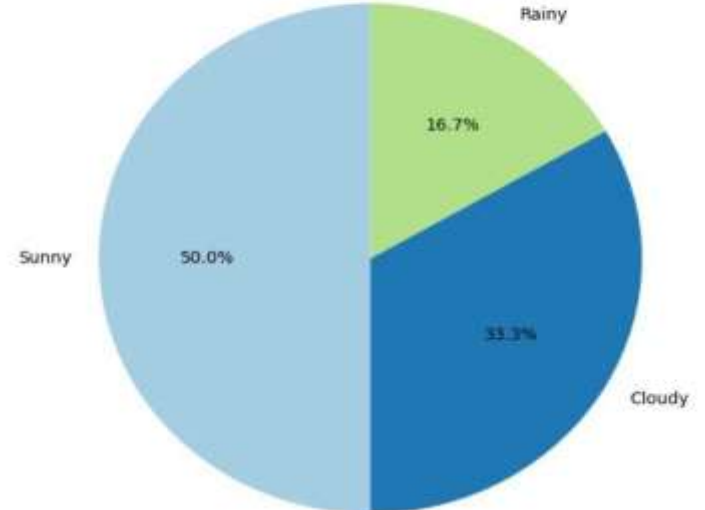
```
weather_descriptions = ["Sunny", "Cloudy", "Rainy", "Sunny", "Cloudy", "Sunny"]  
plot_weather_condition_pie(weather_descriptions)
```



Weather Condition Distribution



```
new_delhi_descriptions = [  
    "Sunny", "cloudy", "Rainy", "Sunny", "Cloudy", "Sunny"  
]  
plot_weather_condition_pie(new_delhi_descriptions)  
plt.title("New Delhi Weather Conditions")
```



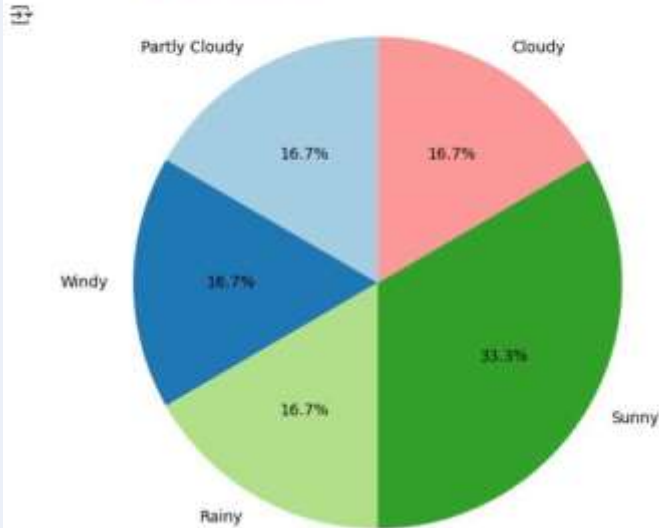


OUTCOMES

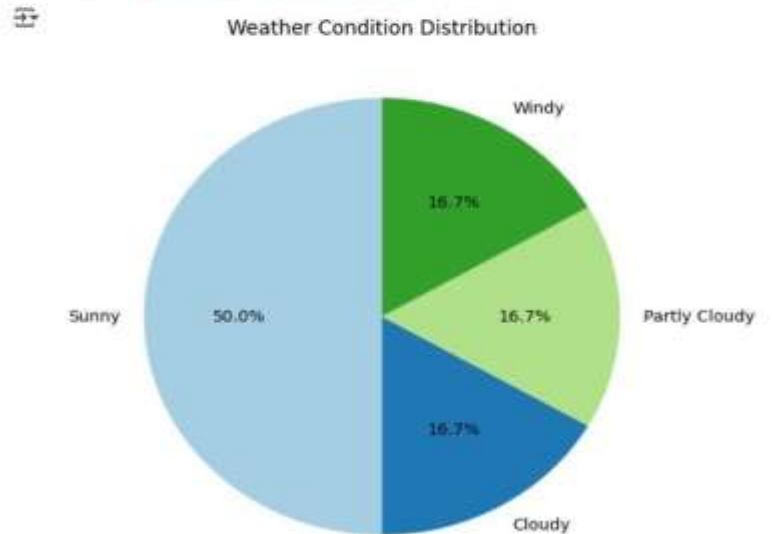


All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025

```
mumbai_descriptions = [  
    "Partly Cloudy", "Windy", "Rainy", "Sunny", "Cloudy", "Sunny"  
]  
plot_weather_condition_pie(mumbai_descriptions)  
plt.title("Mumbai Weather Conditions")
```



```
bangalore_descriptions = [  
    "Sunny", "Cloudy", "Sunny", "Partly cloudy", "Windy", "Sunny"  
]  
plot_weather_condition_pie(bangalore_descriptions)  
plt.title("Bengaluru Weather Conditions")
```





OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025

Enter city name: kolkata

Location: Calcutta, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 34.1°C

Min Temp: 25.8°C

Condition: Rain showers

Date: 2025-04-17

Max Temp: 33.2°C

Min Temp: 26.2°C

Condition: Thunderstorm

Temperature Statistics:

Max Temp Mean: 34.82857142857143°C

Max Temp Median: 34.1°C

Min Temp Mean: 26.085714285714285°C

Min Temp Median: 25.8°C

Weather Condition Frequency:

Rain showers: 2

Thunderstorm: 2

Unknown weather: 1

Partly cloudy: 1

Overcast: 1

Correlation between Max and Min Temperatures: 0.796519490198782

Enter city name: mumbai

Location: Mumbai, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 33.1°C

Min Temp: 26.7°C

Condition: Partly cloudy

Date: 2025-04-17

Max Temp: 32.3°C

Min Temp: 27.7°C

Condition: Partly cloudy

Temperature Statistics:

Max Temp Mean: 33.58571428571428°C

Max Temp Median: 33.1°C

Min Temp Mean: 27.257142857142856°C

Min Temp Median: 27.2°C

Weather Condition Frequency:

Partly cloudy: 7

Correlation between Max and Min Temperatures: -0.22438356214027427

Enter city name: chennai

Location: Chennai, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 35.9°C

Min Temp: 28.0°C

Condition: Rain showers

Date: 2025-04-17

Max Temp: 36.5°C

Min Temp: 27.8°C

Condition: Partly cloudy

Temperature Statistics:

Max Temp Mean: 37.1°C

Max Temp Median: 37.0°C

Min Temp Mean: 28.285714285714285°C

Min Temp Median: 28.4°C

Weather Condition Frequency:

Rain showers: 1

Partly cloudy: 5

Overcast: 1

Correlation between Max and Min Temperatures: 0.7241930560564454



OUTCOMES



All outcomes are based on the data of 16th April 2025 and forecast for 17th April 2025

Enter city name: delhi

Location: Delhi, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 36.9°C

Min Temp: 25.0°C

Condition: Partly cloudy

Date: 2025-04-17

Max Temp: 39.1°C

Min Temp: 27.2°C

Condition: Mainly clear

Temperature Statistics:

Max Temp Mean: 38.1°C

Max Temp Median: 38.3°C

Min Temp Mean: 25.07142857142857°C

Min Temp Median: 25.0°C

Weather Condition Frequency:

Partly cloudy: 1

Mainly clear: 2

Overcast: 2

Clear sky: 2

Correlation between Max and Min Temperatures: -0.1270927780709427

Enter city name: bengaluru

Location: Bengaluru, India

Weather Forecast:

Date: 2025-04-16

Max Temp: 31.4°C

Min Temp: 21.7°C

Condition: Thunderstorm

Date: 2025-04-17

Max Temp: 31.5°C

Min Temp: 20.9°C

Condition: Rain showers

Temperature Statistics:

Max Temp Mean: 32.457142857142856°C

Max Temp Median: 32.8°C

Min Temp Mean: 21.67142857142857°C

Min Temp Median: 21.8°C

Weather Condition Frequency:

Thunderstorm: 2

Rain showers: 2

Unknown weather: 1

Overcast: 2

Correlation between Max and Min Temperatures: 0.43527635984852187



EXTENDED ANALYSIS



•**Dataset Creation:** Compiled a comprehensive dataset encompassing 30 cities from each state of India to ensure diverse geographical representation.

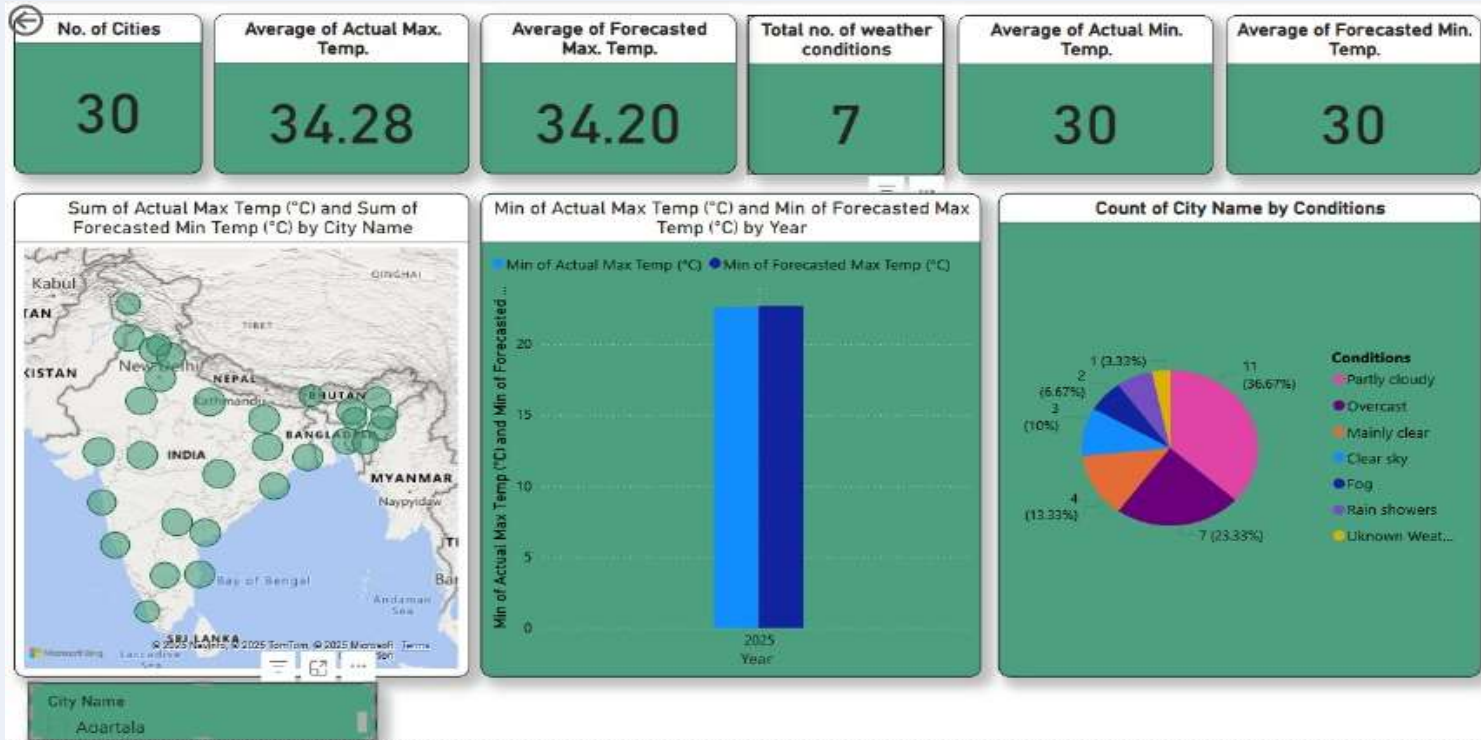
•**Power BI Dashboard:** Utilized Power BI to create an interactive dashboard, summarizing weather patterns across the nation. This visualization aids in identifying regional trends and supports informed decision-making in sectors like agriculture, travel, and event planning.

	A	B	C	D	E	F	G	H	I
1	City Name	Date	Actual Max Temp	Actual Min Temp	Conditions	Forecasted Date	Forecasted Max Temp	Forecasted Min Temp	Forecasted Condition
2	Agartala	24-04-2025	35.7	27.1	Partly cloudy	25-04-2025	34.6	27	Rain showers
3	Ahmedabad	24-04-2025	41.7	25.2	Clear sky	25-04-2025	41	24	Mainly Clear sky
4	Aizawl	24-04-2025	28.2	19.8	Overcast	25-04-2025	27.5	20	Overcast
5	Anantpur	24-04-2025	38.8	19.8	Mainly clear	25-04-2025	39.7	20.2	Overcast
6	Bengaluru	24-04-2025	34.4	22.3	Partly cloudy	25-04-2025	34.3	22.2	Partly cloudy
7	Bhubaneswar	24-04-2025	36.4	28	Overcast	25-04-2025	37.7	27.8	Overcast
8	Chandigarh	24-04-2025	37.6	20.8	Overcast	25-04-2025	38.4	19.4	Overcast
9	Chennai	24-04-2025	37.2	28.6	Partly cloudy	25-04-2025	37.5	29	Partly cloudy
10	Dehradun	24-04-2025	34.5	17.7	Partly cloudy	25-04-2025	35	19	Overcast
11	Delhi	24-04-2025	38.8	23.1	Mainly clear	25-04-2025	38.6	24.7	Mainly Clear sky
12	Gangtok	24-04-2025	22.7	15.1	Rain showers	25-04-2025	22.7	14.4	Rain showers
13	Gunaakati	24-04-2025	36.1	24.3	Partly cloudy	25-04-2025	33.6	24.9	Partly cloudy
14	Hyderabad	24-04-2025	39.2	28.5	Rain showers	25-04-2025	38.7	27.6	Unknown Weather
15	Imphal	24-04-2025	31.3	18.9	Fog	25-04-2025	30.9	19.8	Fog
16	Indore	24-04-2025	39.8	24.5	Mainly clear	25-04-2025	38.6	24.6	Overcast
17	Itanagar	24-04-2025	28.9	21.8	Unknown Weat	25-04-2025	29.4	20.7	Unknown Weather
18	Jaipur	24-04-2025	40.2	24.8	Mainly clear	25-04-2025	40.1	25.8	Overcast
19	Kochi	24-04-2025	23.6	15.6	Overcast	25-04-2025	22.9	12.8	Partly cloudy
20	Kolkata	24-04-2025	37.8	28.4	Overcast	25-04-2025	36.9	28.1	Partly cloudy
21	Kohima	24-04-2025	27.2	18.4	Partly cloudy	25-04-2025	28.1	17.8	Overcast
22	Lucknow	24-04-2025	41.1	24.1	Clear sky	25-04-2025	41.7	25.9	Overcast
23	Mumbai	24-04-2025	32.9	27.7	Overcast	25-04-2025	32.3	27.6	Overcast
24	Panaji	24-04-2025	34.4	29.2	Overcast	25-04-2025	34.7	27.9	Overcast
25	Patna	24-04-2025	40.7	24.9	Clear Sky	25-04-2025	41.4	25.3	Clear sky
26	Raipur	24-04-2025	42.5	29.1	Partly cloudy	25-04-2025	41.7	41.7	Partly cloudy
27	Ranchi	24-04-2025	38.2	26.2	Partly cloudy	25-04-2025	38.3	23.7	Partly cloudy
28	Shillong	24-04-2025	28.8	17.8	Partly cloudy	25-04-2025	28.8	17.8	Partly cloudy

OUTCOMES



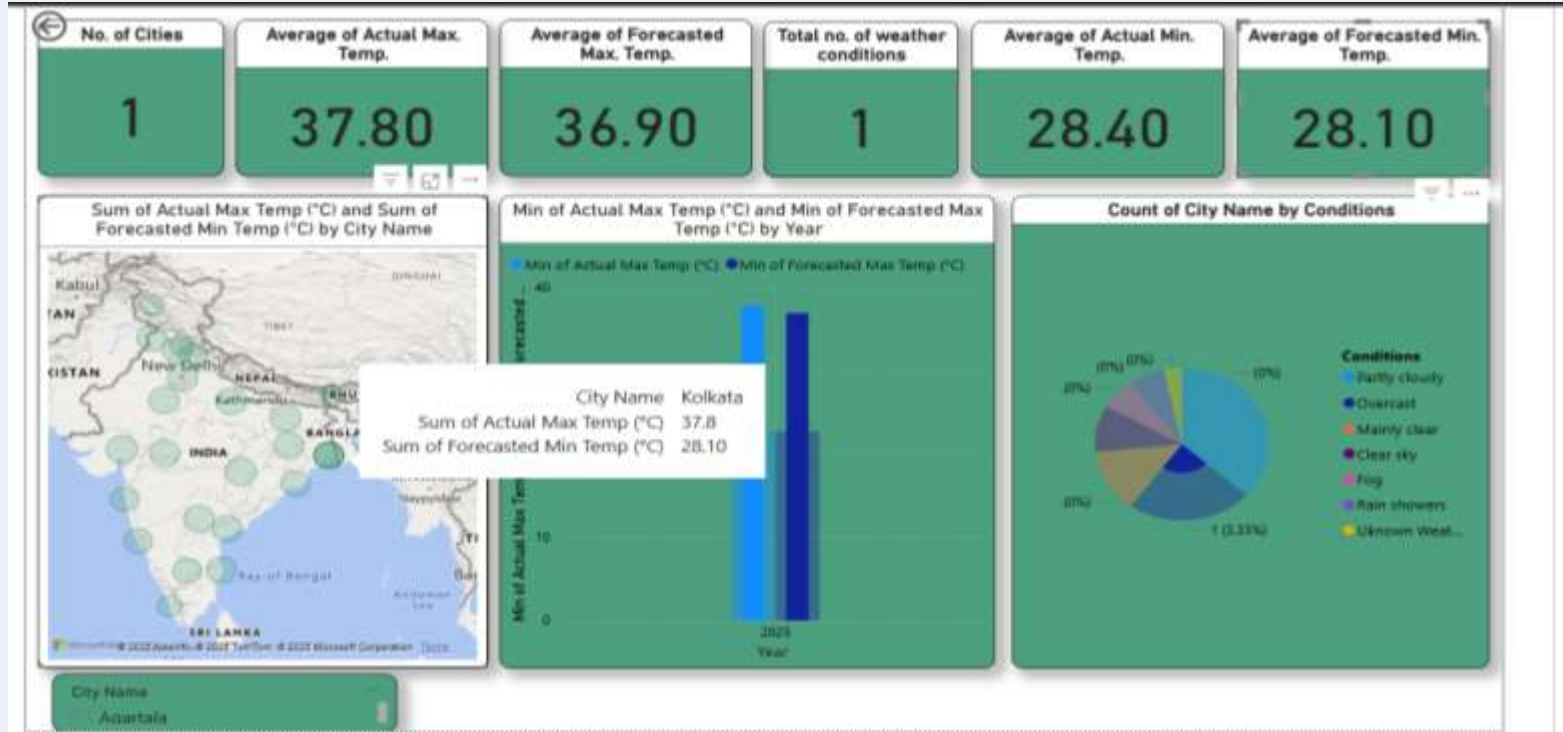
All data are from 24th April, 2025 and forecasted date is 25th April, 2025



OUTCOMES



All data are from 24th April, 2025 and forecasted date is 25th April, 2025





FINDINGS



(All findings are based on the data of 16th April 2025 and forecast for 17th April 2025.)

- Hottest City: Delhi recorded the highest max temperature ($\sim 39^{\circ}\text{C}$).
- Coolest City: Bengaluru showed the lowest max temperature ($\sim 32^{\circ}\text{C}$).
- Most Stable City: Kolkata showed strong consistency (Correlation: 0.79).
- Most Rainy Cities: Kolkata and Bengaluru had frequent Rain Showers and Thunderstorms.
- Most Clear Conditions: Delhi experienced Clear Sky, Mainly Clear, and Overcast.
- Dominant Weather Type: Partly Cloudy was the most common across all cities.
- Statistical Highlights:
 - Strong negative correlation between temperature and humidity.
 - Moderate positive correlation between wind speed and rain.
 - Cloud cover lowers daytime temperatures, especially in monsoon.



FUTURE SCOPE



By analyzing these kinds of outcomes, we can improve and extend the weather forecasting project in the following ways:

- 1.Include More Weather Parameters :** Adding humidity, wind speed, and precipitation forecasts for a more complete weather analysis.
- 2.Apply Machine Learning Techniques :** Building models to predict future weather patterns using historical data and machine learning.
- 3.Create a Public Web Application :** Developing a simple website where users can input a city name and instantly get weather forecasts.
- 4.Use Multiple Weather APIs :** Combining forecasts from different APIs (like OpenWeatherMap, Weatherstack) to increase prediction accuracy.
- 5.Develop Early Warning Systems :** Sending alerts for extreme conditions like heavy rain, heatwaves, or thunderstorms based on forecasts.



CONCLUSION

By completing this project, we can summarize the findings, including weather trends, seasonal patterns, and insights gained from the real-time data analysis. It highlights important factors like temperature fluctuations, dominant weather conditions, and city-wise comparisons. This helps in improving planning for travel, agriculture, event management, and supports better decision-making based on accurate weather predictions.



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